

Card JJ.1: the rollout formulation of the comfort-zone boundary, a design note

2026-09-17, authorized by Jonas the same evening (JJ.Q1: "go as proposed"). Step 1 of the JJ program (docs/active_inference_program.md §10): the construction in full, every parameter with its motivation, and the pre-stated rules of cards JJ.2 to JJ.4. Stop for review before coding: nothing below has been implemented or run, and the rules become binding only when Jonas has read them. Where a value is a placeholder to be replaced by a measured one, it says so. Opinions are marked.

0 What is being built, in one paragraph

At the moment a clip is frozen, roll out a small menu of the ego's options against a fan of the other road user's predicted futures, score every imagined future with the released preference function, and take as the criticality quantity the gap in expected free energy between "continue" and the best alternative, **ΔG , in nats**. Fit the same three-parameter threshold model the project fits on every axis, with **no gate term**, and score it held out against the current best (the gated looming rule) on the second cut-in study, then on the left turn and the cyclist overtake with one level in nats. If ΔG needs a gate to reach the pre-onset cells, or a scenario-specific horizon to reach the left turn, the construction has failed at what it was for and the report says so.

1 The construction

1.0 What the crowd-sourced video data are for in this card

Added 2026-09-18 on Jonas's question. The two video studies are the test bed, not training data. A frozen clip supplies the **scene at the freeze** (positions, speeds, the other's lateral rate, read from the stimulus traces), from which the belief, the fan and the policy rollouts are computed with no response in the loop; the participants supply the **share who would intervene** in that cell. The card then asks one question per cell: does ΔG , computed from the scene alone, order the cells the way the shares do, held out across the design's starting-TTC levels, with the three-parameter threshold model every axis has been given? Nothing in the rollouts is fitted to the responses except the threshold model's three parameters (lapse, level, spread); the predictor's constants come from motivation now and from the generative-model framework later. The pre-onset cells (90 of 378) are the data that test the emergence claim: they are never fitted, only predicted. Study 1's four scenarios play the same role for the transfer (JJ.3), and its 43 drivers seen in every scenario give the per-driver levels for the trait on one scale. Later, on naturalistic data, the response becomes a revealed deceleration and the same ΔG is predicted from the recorded scene, which is card NC.3 of the naturalistic plan run on ΔG .

1.1 The freeze, and the belief about the present

The freeze time t_0 is the end of the shown window: the response moment e_t of the second cut-in study (replication/czb/cutin2_gate.py conventions); the clip end for study 1's cut-in and overtake; the decision moment 13.5 s for the left turn (B3.Q1, src/comfortzone/ltap.py : T_DECISION_S).

The belief $b(s_0)$ is a Gaussian over the two vehicles' positions, speeds and the other's lateral rate, read from the traces exactly as cards G.1 and PC.1 read them: the lateral rate as a backward difference over 0.3 s (second study) or 1.0 s (first study), the same windows those cards used and for the same reason (the stimuli's sample spacing). Its standard deviations are the studies' jitter floors as measured by card HS.1 (replication/czb/out/hs1_situational_surprise.md, read by the script, never typed in). The trap HS.1 fell into, choosing a spread before measuring the floor, is what this step prevents.

A latent intention, for the cut-in only. The belief carries $P(\text{changing lanes}) = p_t$:

- prior p_0 before any lateral motion. Motivation: card G.1's fitted gate sits at 0.063 to 0.070 in every pre-onset cell (out/cutin2_gate.md §4), and the pre-onset intervention share is 0.023 (§2). A prior of **0.07** is the first value; the sweep {0.02, 0.07, 0.20} is reported, and the value is chosen by this motivation, not by the held-out score (§2, rule d).
- update from the lateral rate: a likelihood ratio between "keeping" (lateral rate $\sim N(0, \sigma_{\text{keep}})$, with σ_{keep} the jitter floor of the rate, since G.1 §0 measured the pre-onset rate at -0.010 to 0.003 m/s) and "changing" (lateral rate $\sim N(-v_{lc}, \sigma_{lc})$, with $v_{lc} = 1.2$ m/s, motivated by G.1 §0's post-onset mean of -1.137 m/s and by a 3.65 m lane

crossed in about 3 s; $\sigma_{lc} = 0.4$ m/s, the spread of that same column, to be read from the traces by the script). The posterior after the shown window is p_t . This is the Bayesian ramp of figure 5 in the program document. For the left turn and the overtake there is no intention variable: the oncoming car and the cyclist do what they are seen to do.

1.2 The fan: the predictive distribution over the other's futures

Three predictors, a ladder. JJ.2 to JJ.4 use the first; the second is a secondary in JJ.2; the third waits for JJ.7.

P0, constant-velocity Gaussian with growth. Body-frame lateral position $y(\tau) \sim N(y_0 + \dot{y}_0 \tau, \sigma_y^2 + (\sigma_{v,lat} \tau)^2)$ and longitudinal position $x(\tau) \sim N(x_0 + v_0 \tau, \sigma_x^2 + (\frac{1}{2} \sigma_a \tau^2)^2)$, for τ up to the horizon; under "changing", the lateral rate is v_{lc} toward the ego's lane until the lane center is reached, then held. The mixture weight is p_t . Growth constants, placeholders until JJ.5 measures them on highD (docs/generative_model_framework.md component C3):

constant	first value	motivation	sweep
$\sigma_{v,lat}$	0.33 m/s	G.1's fitted gate spread 0.99 m over its 3 s horizon, read as lateral-rate uncertainty times horizon	{0.1, 0.33, 0.6}
σ_a (longitudinal)	0.5 m/s ²	the pedal-effort tolerance of the released preference is 0.1 m/s ² ; ordinary highway speed variation is larger; unverified , replaced by C3's measurement	{0.25, 0.5, 1.0}
horizon H	6 s	the released model's planning horizon (30 steps of 0.2 s)	{3, 6}
samples per policy	200	enough for a standard error of ΔG below 5% of its cell-to-cell spread (checked in the property tests, not assumed)	—

Common random numbers across policies: every policy is scored against the *same* 200 sampled futures of the other, so ΔG is a difference between paired scores and its Monte Carlo noise is small.

P1, the released norm tournament, open loop. `src/common/dynamics.py::forward_tar_agent` run for 30 steps without observations, with the scenario's norm geometry: the rear-end norms as released, and the crossing norm of `docs/cutin_norm_proposal.md` (`src/comfortzone/norms.py`) as the alternative. This is what the released model would imagine; it is scenario-specific in content (handbook chapter 7) and is reported as a secondary in JJ.2.

P2, a learned predictor (JJ.7): not in this card.

1.3 The policy menu

The menu is the scenario's *instructed alternatives*, which is the one scenario-specific element and is read from each study's instructions rather than chosen:

scenario	policies (the ego's)	notes
cut-in (studies 1 and 2)	continue (hold speed, hold lane); ease off (-1 m/s ²); brake (-3 m/s ²); brake hard (-6 m/s ²)	6 m/s ² is the released model's assumed worst-case lead deceleration; no steering policy (Jonas, 2026-09-18, JJ1.Q1: the crowd-sourced stimuli and the naturalistic data both carry that constraint); steering may be added as a later variant, dated
left turn	proceed (the ego's recorded turn, reading P of card PC.1); wait (stop before the crossing at -3 m/s ²)	the "own planned path" that alone opened PC.1's gate
cyclist overtake	continue the pass at the shown clearance; abort (fall back behind the cyclist at -2 m/s ²)	the overtake's instructed choice

Ego rollouts are deterministic per policy (the ego knows its own plan); all uncertainty is in the fan. The released model's pedal constraints and jerk limits are not modeled: the policies are constant-acceleration segments, and this simplification is recorded.

1.4 Scoring: expected free energy in the released model's own form

For each policy π and each sampled future, the six preference terms of `src/aidriver/preferences.py` are evaluated at every step τ on the imagined observation (relative position, speeds, the ego's own speed, acceleration and lateral position), and

$$G(\pi) = \sum_{\tau} [\max_o \log p(o) - E_{\text{futures}} \log p(o_{\tau} | \pi)] \geq 0,$$

the residual information of the pragmatic value over the horizon, which is exactly the quantity the released model accumulates (Paper Eq. 13; `notes/02_active_inference_overview.md` §2). Two variants, both run in every card because the cost of the second is one flag and one column:

- **Variant A, released:** all six terms, including the safety term's worst-case counterfactual, as the paper has them and as the mirror implements them. Primary, on Jonas's ruling (JJ.Q2, 2026-09-17).
- **Variant B, expected outcome:** the safety term switched off (behind a flag defaulting to the released behavior, rule 3 of the standing rules); collisions are scored where they happen in the sampled futures, by the released collision term with its severity factor. This is the reference the pipeline review pointed at (docs/r2_pipeline_review.md §5). Promoted to primary only by rule (c) of §2.

The desired speed in the speed term is the ego's speed in the clip (the handout's lesson, §4: at the default 15 m/s against an ego at 30 m/s the speed term swamps everything). Epistemic value $\alpha = 0$ in the primary analysis, the authors' validated configuration; $\alpha > 0$ is a secondary, with the caveat `HANDOFF.md` §4 recorded: under looming perception a closer approach sharpens the observation, so epistemic value alone pulls toward approaching.

1.5 The criticality quantity and the response model

$$\Delta G(t_0) = G(\text{continue}) - \min_{\pi} G(\pi) \geq 0.$$

The axis is $\log \Delta G$. If any cell has $\Delta G = 0$ exactly (no sampled future collides under continue, or every alternative costs more control effort than it saves), the report counts those cells and uses $\log(\Delta G + \Delta G_{\min}/2)$ with $\Delta G_{\min}$ the smallest positive value across cells, stated in the report; the sensitivity to that choice is reported as a table.

The response model is unchanged in form: $\text{share who intervene} = \text{lapse} + (1 - \text{lapse}) \times \Phi((\log \Delta G - \log c) / \sigma)$. Three free parameters, the same as every axis competition since gate R.2. **No gate term.** For the per-driver levels (JJ.3) the stage-1 hierarchical estimator is used unchanged with $\log \Delta G$ as its axis (replication/czb/stage1_looming.py as the pattern).

1.6 Where this differs from the falsified field, stated so the difference can be checked

	the field (gate R.2)	ΔG
evaluated on	the present state, pointwise	imagined futures over the horizon
the other's future	one worst case (lead brakes at 6 m/s^2) inside the safety term; one straight line inside the lane-entry gate	a fan under a normative predictor with uncertainty and a latent intention
the ego	the no-action trajectory	a menu; the quantity is relative to the best alternative
the onset	a projection to the moment of closure	the belief about intention
the unit	preference units, one scenario's geometry	nats, every scenario

2 Pre-stated rules for card JJ.2: rollouts on the cut-in (second study)

Cells, folds and metric exactly as the registered gate R.2 script defines them, imported and not modified: 378 cells, 288 post-onset and 90 pre-onset, leave-one-starting-TTC-out folds, weighted RMSE, the chance reference 0.320 and the noise floor 0.118 (out/cutin2_field_vs_gap.md). Comparators, on file: the ungated looming rule 0.1137 (out/cutin2_gate.md §1, model c), the gated looming rule 0.1027 (model k), G.1's pre-onset out-of-sample 0.0319 gated and 0.4832 ungated (§2).

- **(a) Post-onset held-out.** ΔG 's three-parameter threshold model, held out, within **0.01 of the gated looming rule's 0.1027**. The gated rule is the fair comparator because ΔG claims to contain the gate; the ungated 0.1137 is reported beside it.
- **(b) Pre-onset out of sample.** The full post-onset fit applied to the 90 pre-onset cells scores below **0.05**, with no gate term. This is the emergence claim, and it is the rule most likely to fail.
- **(c) The review's ordering constraint.** Within the 24 matched-TTC rows, the sign of ΔG 's rank correlation with the response is positive in at least as many rows as the gap's is negative (docs/r2_pipeline_review.md §4 gives the gap's count). If variant A fails (c) and variant B passes it, variant B is promoted to primary and the report says the released safety counterfactual is what failed, again.
- **(d) Sensitivity, never selection.** p_0 , σ_v , σ_a and H are fixed at the first values of §1 before the run; every sweep value is reported as a table; none is chosen by its score. A reader who prefers a swept value can see what it would have given.
- **(e) Monte Carlo.** ΔG 's standard error across seeds is reported per cell; the verdict is invalid if it exceeds 5% of the between-cell spread of $\log \Delta G$.

Verdict, written before the run: **adopt** ΔG as the primary axis if (a), (b) and (e) hold; **keep as a comparator** if (a) and (e) hold and (b) fails, with the gated looming rule staying primary; **drop** if (a) fails. Rule (c) decides the variant, not the

verdict.

What counts as a failure of the card rather than of the model: a measured jitter floor that makes the lateral-rate likelihood ratio uninformative within the 0.3 s window (report, and fall back to the 1.0 s window with the change dated); any cell whose traces end before t_0 (excluded and counted, as PC.1 did).

3 Pre-stated rules for card JJ.3: transfer in nats (first study)

Left turn (18 cells, `out/ltap_two_axis.md`; distance alone 0.0558 held out, leave-one-PET-out). The perceived arrival of the oncoming car carries the released looming model's distance-dependent noise (handbook chapter 3: constant noise in the angular channel, `decoder.py`), so that the arrival at 70 km/h from farther away is less certain than at 50 km/h.

- **(a) ΔG 's threshold model** within 0.01 of distance alone (0.0558), same folds.
- **(b) Direction of the speed effect.** In the *predicted* shares, the 70 km/h cell is below the 50 km/h cell at every matched PET, as the data are (`docs/lateral_and_uncertainty_note.md` §1). This is the uncertainty mechanism's own prediction, and it is reported whether or not (a) holds.
- **(c) No horizon parameter.** The horizon is the released 6 s; if the "proceed" rollout's conflict at a PET of 4 s (oncoming car arriving 4 s after the ego clears) does not register within it, the report says the emergence claim fails on the left turn at that PET rather than extending the horizon.

Cyclist overtake (15 cells, `out/transfer_overtake_summary.md`).

- **(d) Intervention** graded in clearance in the predicted shares across the three clearance levels, which the field could not produce (`out/overtake_field_check.md`), with the held-out score reported against the lateral-clearance rule of card B.1.

The trait on one scale. Per-driver levels on log ΔG fitted on the cut-in and on the left turn for the 43 drivers seen in both (card TR.1's set, `out/driver_levels.md`): Spearman correlation with its bootstrap interval, against TR.1's +0.647 in mixed units and against the reliability ceiling. Pre-stated reading: a correlation within the interval of TR.1's is "the same trait on one scale, no rescaling"; a correlation clearly above it would be the first evidence that the common unit adds information; below it, that ΔG loses per-driver signal that the scenario-specific axes keep. **EL.Q4 is answered by this card only if the first or second reading obtains.**

4 Pre-stated rules for card JJ.4: precision as spread

Two refits of the stage-1 estimator, each replacing free spreads by a precision tied to the stimulus:

- **Medium.** One spread for the video left turn and one for the track (card TT.1's data, `external/02_LTAP0D_DBIN/`), the ratio free, against the two free spreads already on file (0.86 s and 0.20 s). Trivially equivalent as a fit; the content is the second step: the ratio predicted from the perceptual model (angular noise on a monocular frozen frame against a moving binocular view) is reported beside the fitted ratio. **Unverified constants** for the perceptual side are marked and sourced in the script or the step is reported as not yet possible.
- **Apparent size.** The truck and car cut-ins of study 1 at matched TTC, the roadmap's E.2: a model with one spread per stimulus class against one shared spread. Rule: the two-spread model earns its place if it improves held-out log-likelihood by more than 2 units under LOPO; the direction (truck sharper) is reported either way.
- **The gate's spread, derived.** G.1's $s_l = 0.990$ m against $\sigma_{v,lat} \times 3$ s with $\sigma_{v,lat}$ measured from the traces' lateral-rate jitter plus P0's growth constant; reported as a computed value beside the fitted one. Agreement within a factor of 1.5 is the pre-stated reading for "the gate's spread is a computed uncertainty".

5 Software plan

New package `src/rollout/`, nothing in `src/aidriver/preferences.py` changed (variant B is a flag defaulting to the released behavior):

module	contents	property tests (<code>tests/test_rollout.py</code> , <code>check()</code> style)
<code>belief.py</code>	the Gaussian belief at the freeze; the intention posterior	prior recovered when the rate is at the jitter floor; posterior $\rightarrow 1$ for a rate at $-v_{lc}$ sustained over the window; symmetric under sign convention
<code>predictor.py</code>	P0 with growth and the intention mixture; P1 as a wrapper on <code>forward_tar_agent</code>	zero growth reproduces G.1's projection exactly (the step limit); the sample mean matches the analytic mean to Monte Carlo error; common random numbers give identical fans across policies

module	contents	property tests (tests/test_rollout.py, check() style)
policies.py	the menus; constant-acceleration rollouts	continue holds speed; brake hard reaches 0 at v/6 s; the left-turn "proceed" reproduces PC.1's reading P path to within its residual
efe.py	G(π) from preferences.py terms; variants A and B; α	$G \geq 0$; identical policies give identical G; a certain collision under continue and none under brake gives $\Delta G > 0$ growing as the gap shrinks; with $p_0 = 0$ and a lane-keeping other, $\Delta G = 0$
boundary.py	ΔG per cell; log axis with the zero rule; the standard error across seeds	the zero rule applies only when a zero exists; SE reported

Scripts, one per card, in replication/czb/: je2_rollout_cutin.py (imports the registered R.2 script's cell construction, folds and metric; writes out/je2_rollout_cutin.md plus per-cell CSV), je3_rollout_transfer.py ($\rightarrow$ out/je3_rollout_transfer.md), je4_precision_spread.py ($\rightarrow$ out/je4_precision_spread.md). Each writes its main report before any bootstrap (rule 10 of the handover). The suite grows by one file, tests/test_rollout.py, and the handover's §0 list is updated.

Cost, in compute: 378 cells $\times$ 4 policies $\times$ 200 futures $\times$ 30 steps of six preference terms is of the order of 10^7 term evaluations, seconds in NumPy. In work: the package and tests two to three days; JJ.2 one day of runs and writing; JJ.3 two days (the left-turn and overtake menus and the looming noise on the arrival); JJ.4 one day.

6 What could go wrong, stated before the run

- The released preference costs are enormous (collision $-10\,000$, road edge $-15\,000$); ΔG in nats will be in the thousands and log ΔG 's spread across cells may be dominated by how many sampled futures collide. That is the intended mechanism, but it means the Monte Carlo rule (e) is not a formality.
- Variant A may pass (a) and fail (c): the released safety term inside the rollouts would then order by absolute speed as it did pointwise. The card's design anticipates this with variant B.
- The intention update may be too slow within 0.3 s of shown lateral motion to lift ΔG in the earliest post-onset cells (CP2), in which case ΔG behaves like the ungated rule there; the report gives the pre- and post-onset fits separately so this is visible.
- On the left turn, the released looming noise may be too small at 60 m to grade the arrival, in which case rule 3(b) fails for a reason that is about the noise constant, not the mechanism; the report says which.
- The passenger's alternatives in the video may not be the menu's; a scenario offset would then be absorbed into the level and is not separable within this design.

7 Queries, and Jonas's rulings of 2026-09-18

- **JJ1.Q1, resolved:** no steering policy in the menus we start with; the crowd-sourced stimuli and the naturalistic data both carry that constraint. Section 1.3 updated.
- JJ1.Q2 (minor, review): $\sigma_a = 0.5\text{ m/s}^2$ for longitudinal growth is unverified and is the constant the generative-model framework's component C3 replaces. Until then it is a placeholder with a sweep; flagged so no report quotes it as measured.
- **JJ1.Q3, resolved as proposed:** JJ.4 reports the fitted ratio of the video and track spreads, and marks the ratio *predicted from perception* as not yet possible unless a source for the perceptual constants is found. In plain words: we can measure that drivers are four times sharper in the car; to *predict* that number from how much less precisely a frozen monocular clip conveys distance and closing speed, we would need published values of that visual precision, and we do not have them.
- **Authorization (2026-09-18): "go."** Implementation is delegated to a less expensive model; the brief for it is handover_jj1_implementation.md, which carries every reference, signature and rule the implementer needs without reading this note's argument.